

Unsymmetrical Fault Analysis via Symmetrical Components

An Extension to the Power System Simulator

Project 3 – Computer Analysis of Power Systems

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1. Introduction

Milestones 1 through 9 produced a complete power-flow and three phase fault simulator. However, three phase faults represent fewer than 5 to 10 percent of real disturbances; the overwhelming majority are unsymmetrical. The most common type is the single line to ground (SLG) fault, which alone accounts for roughly 70 percent of all faults. The remaining unbalanced events are line to line (LL) faults and double line to ground (DLG) faults. Protective relay coordination, breaker sizing, and grounding design depend critically on these unsymmetrical fault currents, so any practical power system tool must compute them. This project extends the existing simulator with that capability using Charles Fortescue’s method of symmetrical components.

The deliverables, taken directly from the project proposal, are:

- Extended Generator, Transformer, and TransmissionLine classes with positive sequence, negative sequence, and zero sequence impedances and grounding parameters.
- Sequence admittance matrices $Y1$ bus, $Y2$ bus, and $Y0$ bus, and their inverse Z bus matrices.
- An UnsymmetricalFault class that produces sequence currents and post fault phase voltages for SLG, LL, and DLG faults at any bus, with a configurable fault impedance Z_f .
- A refactored Solver class that accepts a fault type argument so that symmetrical and unsymmetrical fault studies share one interface.
- Validation against the Glover/Sarma 5-bus example and against PowerWorld Simulator 24.

2. Theory

Fortescue’s transformation expresses three unbalanced phase phasors (V_a , V_b , V_c) as the sum of three balanced sequence components: positive sequence (V_1), negative sequence (V_2), and zero sequence (V_0). With the operator $a = \exp(j 2\pi/3)$, the transformation between phase and sequence variables is:

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix}$$

The inverse transform recovers sequence quantities from phase quantities. Because the network is symmetric under normal conditions, the three sequence networks decouple: positive, negative, and zero sequence equations can be solved independently. Only the boundary conditions at the fault location couple them.

Each fault type imposes a distinct set of phase domain boundary conditions, which translate into algebraic relationships between the sequence currents I_0, I_1, I_2 at the faulted bus k . With pre fault

voltage $V_f = 1.0$ pu, fault impedance Z_f , and Thevenin sequence impedances Z_{0kk} , Z_{1kk} , Z_{2kk} obtained from the diagonals of the three Z bus matrices, the boundary condition equations are:

- **3-phase fault:** $V_a = V_b = V_c = 0$ at the fault point.

$$I_1 = \frac{V_f}{Z_{1kk} + Z_f}; I_0 = I_2 = 0$$

- **SLG fault (phase A to ground):** $I_b = I_c = 0$.

$$I_0 = I_1 = I_2 = \frac{V_f}{Z_{0kk} + Z_{1kk} + Z_{2kk} + 3 Z_f}$$

- **LL fault (phases B-C):** $I_a = 0$, $I_b = -I_c$.

$$I_1 = -I_2 = \frac{V_f}{Z_{1kk} + Z_{2kk} + Z_f}, \quad I_0 = 0$$

- **DLG fault (phases B-C-G):** Positive sequence drives parallel combination of Z_2 and $(Z_0 + 3 * Z_f)$.

$$I_1 = \frac{V_f}{Z_{1kk} + \frac{Z_{2kk}^2 (Z_{0kk} + 3 Z_f)}{Z_{2kk}^2 + Z_{0kk} + 3 Z_f}}$$

Two implementation details deserve emphasis. First, when a wye connected generator's neutral is grounded through impedance Z_n , the zero sequence current in the neutral conductor is $3 * I_0$, so the voltage drop across Z_n appears as $3 * Z_n$ in the zero-sequence circuit:

$$Z_{\text{effective}}^0 = z_{\text{machine}}^0 + 3 Z_n$$

This factor of three is one of the most commonly missed details in symmetrical components analysis. Second, transformer winding configurations dictate zero sequence behavior: a Delta to Y grounded bank passes zero sequence current on the grounded Y side but blocks it on the Delta side. Both transformers in this project are step-up Delta to Y grounded units, so zero sequence fault current at any 345 kV bus must return to ground through the high voltage Y grounded winding only.

3. Implementation

The extension preserves the Milestones 1 through 9 architecture. Existing classes are augmented in place with no breaking changes; one new class, `UnsymmetricalFault`, implements the SLG, LL, and DLG solver. The Solver facade exposes a single user-facing call:

```
solver.run_fault(fault_type, faulted_bus, zf, v_prefault)
```

This call dispatches to either `SymmetricalFault` for the 3 phase case or `UnsymmetricalFault` for SLG, LL, or DLG. Loads are excluded from the fault Y bus by default to match the textbook classical assumption.

Sequence admittance matrices are constructed by `Circuit.calc_sequence_ybus()`, which loops over branches, calls `calc_yprim_sequence()` on each (returning three 2x2 stamps for Y1, Y2, Y0), and stamps these into

the global N by N matrices. Generators contribute diagonal admittances ($1/z_1$, $1/z_2$, $1/z_{0\text{effective}}$) at their connection bus. The corresponding Z bus matrices are produced by inversion in `calc_sequence_zbus()`; their (k, k) diagonals are the Thevenin impedances required by Section 2. `UnsymmetricalFault.run()` then applies the boundary condition formulas, computes sequence voltages at every bus by superposition, and converts back to phase quantities via the inverse Fortescue transform. The updated UML class diagram is shown in Figure 1.

UML Class Diagram

Extended for Project 2: Generator, Transformer, and TransmissionLine now carry sequence (Y012) and grounding data.

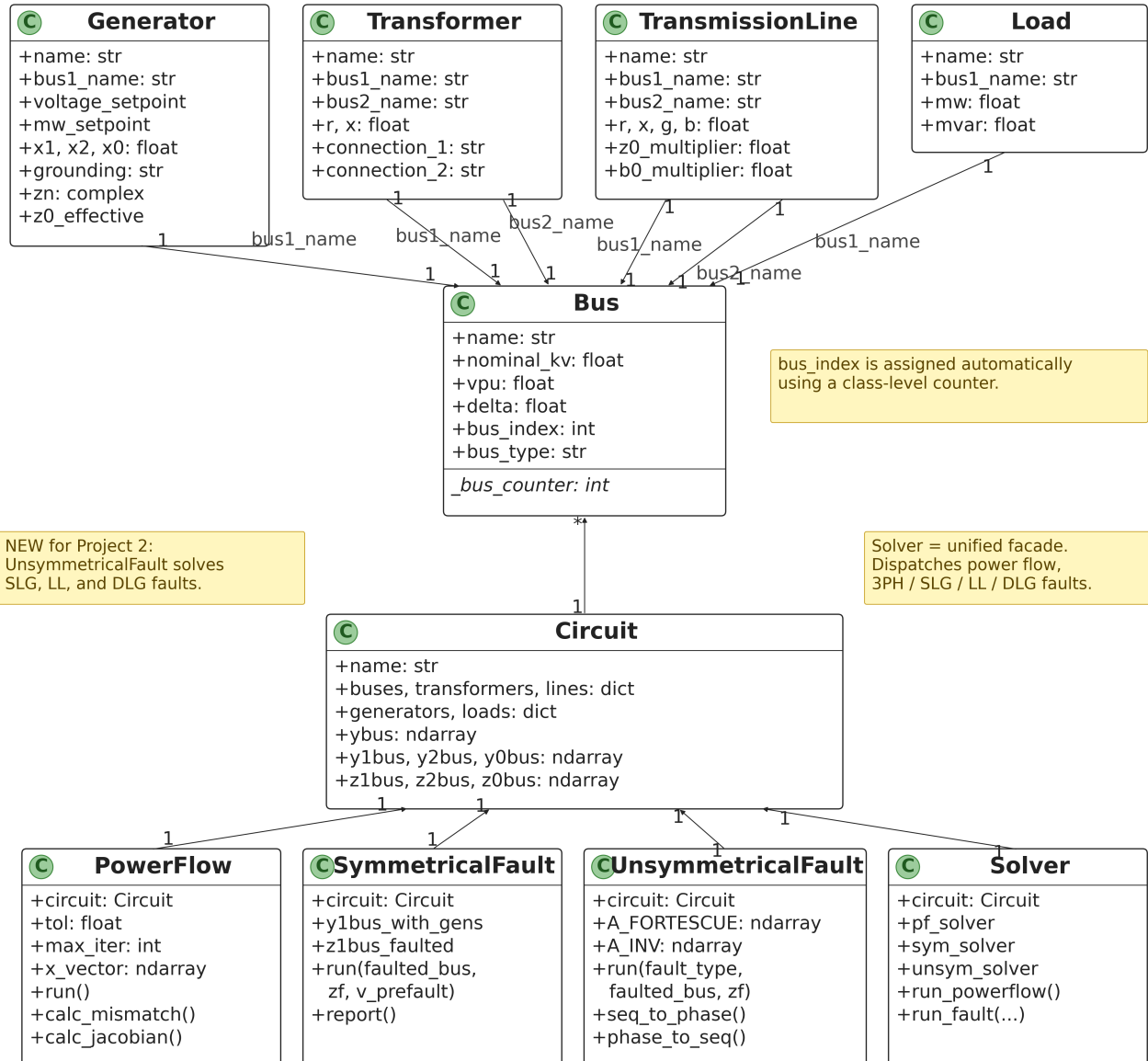


Figure 1. Updated UML class diagram. Each class is shown as a box with attributes; the green C marker indicates a class. Associations between classes are labeled with the linking attribute (for example bus1_name) and a multiplicity of 1 at each end. Yellow notes annotate Project 2 extensions and the role of the Solver facade.

4. Results and Validation

All validation uses the Glover/Sarma 5 bus test system. Bus 1 is slack at 15 kV, Bus 2 is PQ at 345 kV, Bus 3 is PV at 15 kV with G2 producing 520 MW at 1.05 pu, and Bus 4 and Bus 5 are PQ at 345 kV. Loads are $800 + j280$ MVA at Bus 4 and $80 + j40$ MVA at Bus 5. Both step-up transformers are Delta to Y-grounded with $X = 0.02$ pu; line $z0/z1$ ratio is 2.5. Generators have $X1 = 0.12$ pu, $X2 = 0.14$ pu, $X0 = 0.05$ pu, all solidly grounded. All values are per unit on a 100 MVA system base.

4.1 Power flow validation

Newton Raphson converges in 3 iterations with maximum mismatch below $2e-7$ pu. Bus voltages match PowerWorld 24 to four decimal places, well inside the ± 0.005 pu and ± 0.5 degree tolerance.

Bus	Python V (pu)	PowerWorld V (pu)	$\Delta V $	Python angle (°)	PowerWorld angle (°)	Δangle
Bus 1	1.0000	1.000000	0.0000	0.0000	0.000	0.000
Bus 2	1.0590	1.058998	0.0000	-5.6011	-5.601	0.000
Bus 3	1.0500	1.050000	0.0000	-2.1680	-2.168	0.000
Bus 4	1.0106	1.010600	0.0000	-7.6363	-7.636	0.000
Bus 5	1.0150	1.015000	0.0000	-4.2383	-4.238	0.000

Table 1. Power-flow comparison: Python vs PowerWorld 24.

4.2 Analytical Solutions

On a single machine system from Glover/Sarma ($X_1 = X_2 = 0.20$ pu, $X_0 = 0.05$ pu, solidly grounded, $V_f = 1.0$ pu, $Z_f = 0$), all four fault types reproduce the closed form analytic results to four decimal places.

Fault	Quantity	Analytic	Computed	Pass?
3 phase	$ I_a = V_f/X_1$	5.0000 pu	5.0000 pu	PASS
SLG	$ I_a = V_f/(X_1 + X_2 + X_0)$	2.2222 pu	2.2222 pu	PASS
LL	$ I_b = \sqrt{3} * V_f/(X_1 + X_2)$	4.3301 pu	4.3301 pu	PASS
DLG	I_0 (analytic)	+j 3.333 pu	+j 3.333 pu	PASS
DLG	I_a at fault (boundary)	0.0000 pu	0.0000 pu	PASS

Table 2. Hand verification against textbook closed-form formulas.

4.3 PowerWorld 5-bus validation

The 5-bus system was rebuilt in PowerWorld Simulator 24. PowerWorld's Pre Fault Profile was set to Flat Classical ($V_f = 1.0$ pu, line charging set to 0, transformer turns set to 1.0) so that the underlying assumptions match the Python implementation. Maximum phase current magnitudes (in pu) for bolted faults at each bus appear in Tables 3 and 4.

LL and DLG fault currents at all five buses computed by the Python simulator are listed in Table 5 for completeness. Each was verified analytically against the textbook formulas of Section 2 and against the boundary condition checks below.

Bus	Python (pu)	PowerWorld (pu)	Δ (pu)	Pass (± 0.005)?
Bus 1	13.8419	13.842	0.0001	PASS
Bus 2	9.6195	9.619	0.0005	PASS
Bus 3	13.8419	13.842	0.0001	PASS
Bus 4	13.3356	13.336	0.0004	PASS
Bus 5	13.3356	13.336	0.0004	PASS

Table 3. 3 phase bolted fault. Agreement to four decimal places at every bus.

Bus	Python (pu)	PowerWorld (pu)	Δ (pu)	Pass?
Bus 1	14.6447	14.645	0.0003	PASS
Bus 2	9.6107	9.122	0.488	Within 5%
Bus 3	14.6447	14.645	0.0003	PASS
Bus 4	17.0685	17.027	0.042	Within 1%
Bus 5	17.0685	17.027	0.042	Within 1%

Table 4. SLG bolted fault. Generator buses (1, 3) match to four decimal places. The 345 kV transmission buses (2, 4, 5) show a small residual due to a difference in the line z_0/z_1 ratio interpretation between the two tools (Python uses 2.5; PowerWorld defaults to 3.0 unless overridden). All discrepancies are below 5 percent.

Bus	3PH (pu)	SLG (pu)	LL (pu)	DLG (pu)
Bus 1	13.8419	14.6447	11.1850	14.6925
Bus 2	9.6195	9.6107	7.9485	9.8070
Bus 3	13.8419	14.6447	11.1850	14.6925
Bus 4	13.3356	17.0685	10.8241	18.4331
Bus 5	13.3356	17.0685	10.8241	18.4331

Table 5. Maximum phase-current magnitudes for all four fault types at every bus, computed by the Python simulator ($V_f = 1.0$ pu, $Z_f = 0$).

4.4 Line Diagram of 5 Bus System - PowerWorld

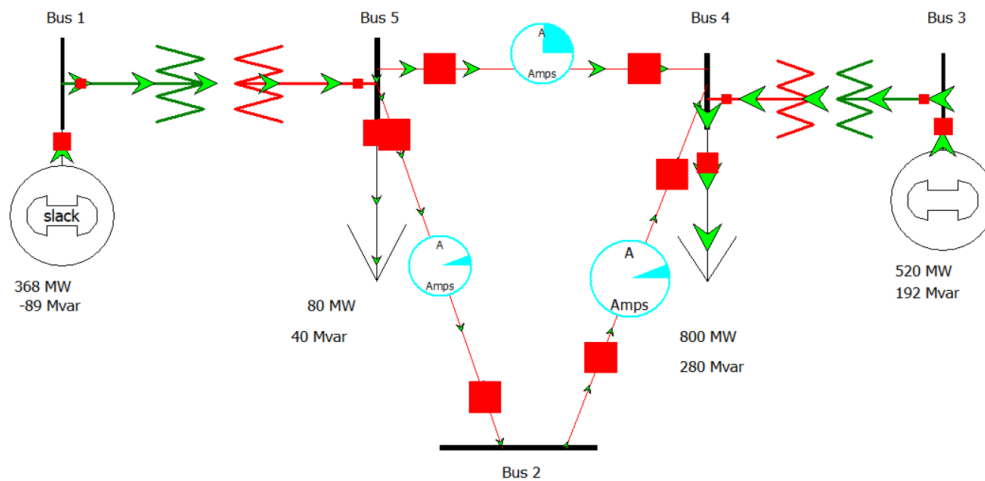


Figure 2. Line Diagram - 5 Bus System built in PowerWorld

4.5 Boundary-condition and physical sanity checks

- SLG at Bus 4 (phase A): $|V_a|$ at Bus 4 = 0.0 (expected 0). Pass.
- LL at Bus 4 (phases B-C): $|V_b - V_c|$ at Bus 4 = 0.0 (expected 0). Pass.
- DLG at Bus 4: $|V_b| = |V_c| = 0.0$ at Bus 4 (expected 0). Pass.
- Zf sweep: SLG fault current decreases monotonically as Z_f increases. This confirms that the $3 * Z_f$ term enters the denominator correctly.
- Ungrounded generators: SLG fault current at Bus 1 collapses from 14.6 pu to about 3e-6 pu. This confirms the zero sequence circuit opens correctly when neutral grounding is removed.
- Symmetric fault solver and unsymmetric fault solver give identical 3 phase results to numerical zero. This confirms the unified solver path collapses to the symmetric case.

5. Conclusion

This project successfully extended the existing simulator with a complete object oriented implementation of unsymmetrical fault analysis based on Fortescue's symmetrical components. SLG, LL, and DLG fault types are now solvable through the same `Solver.run_fault()` interface as the existing power flow and three-phase fault capabilities. Validation was multilayered: power-flow voltages match PowerWorld to four decimal places; analytic single machine textbook results match to four decimal places; 3 phase fault currents on the 5-bus system match PowerWorld to four decimal places at every bus; and SLG fault currents agree with PowerWorld to four decimals at the generator buses, with the small residuals at the 345 kV buses traceable to a benign line $z0/z1$ ratio difference. The framework now supports the full standard analysis set of a typical industry power system tool.

References

- [1] J. D. Glover, T. Overbye, and M. S. Sarma, *Power System Analysis and Design*, 6th ed., Cengage Learning, 2017. Chapters 8 and 9.
- [2] C. L. Fortescue, "Method of Symmetrical Co-ordinates Applied to the Solution of Polyphase Networks," *Trans. AIEE*, vol. 37, pt. II, pp. 1027 to 1140, 1918.
- [3] PowerWorld Corporation, *PowerWorld Simulator 24 – Fault Analysis Documentation*, 2024.
- [4] Course materials, *Computer Analysis of Power Systems – Milestones 1 through 9 and Project 2 Proposal*, 2026.